

Santhosh Bharadwaj H A

Senior Software Engineer | 5+ Years of Experience

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PROFESSIONAL SUMMARY

- Senior Software Engineer with 5+ years of experience designing and building scalable distributed systems, data platforms, and cloud-native applications using Python, PySpark, SQL, and AWS.
- Proven expertise in enterprise AI and data solutions, including multi-agent conversational AI platforms, GenAI-powered document processing pipelines, and high-volume data migration systems handling millions of records.
- Strong proficiency in serverless architecture, ETL and ELT pipelines, workflow orchestration, and data warehousing using AWS Lambda, Glue, S3, DynamoDB, Step Functions, Redshift, Bedrock, and CDK.
- Demonstrated success in improving system performance, data quality, observability, and operational reliability through optimized pipelines, reconciliation frameworks, and scalable backend services.
- Adept at solving complex engineering problems and partnering with cross-functional teams to deliver high-impact, production-ready solutions across telecom, healthcare, and analytics domains.

EXPERIENCE

Senior Software Engineer | Prodapt Solutions Pvt. Ltd.

May 2025 - Present

1. AI Voice Bot Platform for Caribbean Telecom

Description: Designed and developed a multi-agent conversational AI platform for 14 Caribbean markets, supporting billing, authentication, diagnostics, and customer support across IVR, WhatsApp, and app channels. Built with AWS Bedrock AgentCore, Strands Agents, DynamoDB, Lambda, and CDK for scalable, stateful, and observable enterprise deployment.

2. Large-Scale Customer Data Migration (18M+ Records)

Description: Designed and implemented a high-volume customer data migration platform to move and validate 18M+ records across multiple source and target databases. Built ETL, reconciliation, performance optimization, and monitoring workflows using PySpark, AWS Glue, S3, Lambda, and Step Functions.

Software Engineer | Medilenz Innovation Private Limited

Jan 2024 - May 2025

1. GenAI-Powered Medical Record Review Pipeline

Description: Designed and developed an end-to-end pipeline for large-scale processing of unstructured medical records. Built an OCR to structuring to analysis workflow to normalize data, store outputs in Amazon S3 (CSV), and use Amazon Bedrock for clinical summarization, insight generation, and downstream review support.

Data Engineer | SMS TechSoft

Feb 2021 - Dec 2024

1. Data Lake Automation with AWS Glue for Sales Optimization

Description: Designed and developed an automated AWS data lake solution using Python, Amazon S3, AWS Glue, and Amazon Redshift for scalable data ingestion, transformation, and querying. Enabled reliable analytics and data-driven decision-making to support sales optimization.

2. Personalized Learning Pathways with Real-Time Feedback

Description: Developed and maintained an end-to-end data pipeline using Python, Apache Airflow, and Amazon Redshift to extract, clean, transform, and integrate data from multiple sources. Delivered high-quality, near real-time data to support personalized learning experiences and feedback systems.

SKILLS

Programming

Python, SQL, NoSQL, Graph DB, PySpark, Pandas

Cloud & AWS

AWS, Amazon S3, AWS Lambda, AWS Glue, AWS Step Functions, Amazon Redshift, Amazon DynamoDB, API Gateway, AWS Bedrock, AWS CDK, Secrets Manager, CloudWatch, AWS X-Ray, Cloud Trails, IAM, Code Artifact

AI / GenAI

Generative AI, Agentic AI, Multi-Agent Systems, Amazon Bedrock, AWS AgentCore, Strands Agents, Claude Haiku, Prompt Orchestration, Conversational AI, NLP, OCR, Clinical Summarization

Data Engineering	ETL/ELT Pipelines, Data Migration, Data Integration, Data Transformation, Data Validation, Data Cleansing, Data Quality, Data Reconciliation, Schema Design, Data Warehousing, Real-Time / Near Real-Time Processing
Databases	PostgreSQL, MySQL, MariaDB, MongoDB, Oracle, Amazon Redshift, DynamoDB
APIs & Backend	RESTful APIs, FastAPI, Django, Flask, WebSocket, Model Context Protocol (MCP)
Workflow	Apache Airflow, Glue Workflows, Step Functions, YAML-based Flow Design, Async Processing
DevOps & Tools	Git, GitHub Actions, Docker, CI/CD, OpenTelemetry, Structured Logging, Monitoring & Observability
Core Engineering	Distributed Systems, Serverless Architecture, Performance Optimization, Query Optimization, Scalability, Debugging, Cross-Functional Collaboration

EDUCATION

B.E. in Information Science & Engineering	2017 - 2021
Visvesvaraya Technological University	
Diploma in Computer Application (DCA)	Graduated: June 2017
Shri Swami Vivekananda Institute	